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Project Description

This academic year, our senior design project will focus on developing a solution to an open-ended problem provided by OWL Integrations (OWL). In the past, OWL has sponsored many projects, including but not limited to an AI-powered shark detection system deployed in the Long Island Sound, and LoRa network nodes that were launched and communicated with in near orbit. This time, OWL provided us with their Internet-of-Things (IoT) hardware and supplemental software to solve a real-world problem, whether it is personal or at UConn. As such, we will collaborate with OWL and the University of Connecticut's (UConn) ECE department to develop a real-time data monitoring solution for the UConn Formula SAE (FSAE) team's electric racecar. Currently, FSAE does not have the capabilities to monitor the vehicle's status while it is on the racetrack. Our solution aims to create a system that collects real-time data from the vehicle to be transmitted to the FSAE pit crew. From which, they would be able to view relevant data and real-time updates on the racecar on a web-based dashboard. This new capability would allow the pit crew to make informed decisions regarding the vehicle's performance and perform necessary repairs. This system will also detect anomalies within the vehicle itself, primarily within the vehicle's battery.

As mentioned before, OWL will be providing us with the necessary hardware and supplemental software to aid in the development of this project. As they are specialized in ad hoc

mesh network solutions, OWL offers "Quackerboard" ("ducks") hardware, which acts as nodes in the mesh network, and their "Duck Management System" (DMS) software, an easy-to-set-up cloud platform for monitoring the ducks in the network. The ducks communicate with each other and to the DMS using the "ClusterDuck Protocol," which is based on the long-range, low-power transmission standard of LoRa ("Empowering Communities through Connectivity", 2025). While the transfer rate of LoRa is significantly less than the likes of 5G and other cellular transmissions ("Lora vs. 5G for IOT Applications: Which Is Best for You?", 2025), LoRa does not require a license to use, and it is free ("Lorawan", 2025). We will use these to create the system of data acquisition, processing, and visualization. By the end of the academic year, this system will be complete, with complete documentation on deployment.

This project offers an exploration of new ideas and concepts. It will serve as an excellent exercise regarding machine learning, web development, and embedded systems. In addition, developing something that runs on physical, bare metal systems is something I am very interested in. The networking part of the project of data transfer and transmission protocols is also what piqued my interest. This interest is not without its hurdles, however. We will need to ensure that the machine learning model is computationally and energy efficient, and that it has been sufficiently trained to detect anomalies. We will also need to ensure that the transmission of data is fast with little latency and secure.

On the CSE side, this project is split into two parts. As mentioned before, these parts are data acquisition, processing, and visualization. Data acquisition will be done on the car with the help of sensors attached to it. The data collected from the sensors is then transmitted to an intermediate device for processing via the CAN network. The processed data will then be aggregated to the Quackerboard and transmitted via LoRa. The ECE department will make sure

that the sensors that interface with the vehicle work nominally. In addition, we will be training a machine learning model to detect anomalies within the vehicle, based on the data samples collected by the sensors. Detected anomalies and collected data are then sent to the other part of the project, the web dashboard. In essence, the ECE department will be responsible for data acquisition. Data processing and detecting anomalies will be the machine learning side of the project, and the data visualization will be the web development side of the project. Since the effort of this project is split into two, team members Jobe Goetz and Maeve Smith will work on the machine learning side, and Esha Ramireddy Gari and I will work on the web development side.

There are two ways to implement the transfer of data from the DMS to the dashboard. Currently, with how OWL has developed the interaction between the ducks and the DMS, all the ducks in the mesh network would report their data to a network gateway, which would be sent to the cloud; in this case, the DMS (Knouse, 2024). As such, data would be retrieved by means of RESTful API calls from the dashboard to the DMS. According to OWL's lead software architect, Charlie Evans, API calls to the DMS can be done with any language and framework we are comfortable with. This rules out the need to choose a "fast" framework; we are considering using the data visualization library called Plotly in Python. However, it is important to remember that the data from the ducks must be transmitted to the network gateway and then to the DMS before being retrieved by the dashboard. This raises the concern for data transfer latency. The second way to implement the transfer addresses this concern, in that the data would be transferred through a WebSocket, which allows us to get the data in real-time. However, OWL currently does not have support for WebSockets for the DMS at the moment. If latency becomes a big problem, then WebSockets must be considered and implemented. In any case, the data would be

retrieved from the DMS and displayed in real-time, with dynamic visuals to indicate parameter changes and anomaly notifications. The dashboard will also include post-race diagnostics and analysis, with graphs of relevant parameters plotted over the course of the race.

As mentioned above, Jobe Goetz and Maeve Smith are working on the battery anomaly detection model. After some research, they have determined that the model will be initially based on the Isolation Forest model, with plans next semester to develop a Long-Short-Term-Memory (LSTM) Autoencoder model. The reason for considering the LSTM Autoencoder is that its training is unsupervised, meaning that it will detect irregularities given a normal training dataset; irregularities occur if reconstruction error is high (Hong, 2024). However, training such a model requires a magnitude of computational resources that we currently do not have access to. For this reason, we are considering training an Isolation Forest model instead, which is an efficient and simple algorithm for anomaly detection ("What Is Isolation Forest?", 2025). FSAE has provided us with such data sets from their testing to train our models on. Once trained, the model will then be put on Google's Coral Board, a board dedicated for running models on the edge; "a single-board computer that contains an Edge TPU coprocessor" ("Get Started with the Dev Board", 2025). There are concerns about optimal performance for this type of edge computing. For example, is the model enough for the Coral Board, or can the model correctly detect actual anomalies?

As for collaboration, our team will be working under the agile scrum framework, incorporating sprints via Jira to keep track of what is next in our project development on a week-by-week basis. As mentioned above, our project is split into two parts, with Jobe and Maeve on machine learning, and Esha and I on web development. We meet each week, online and in-person, to discuss our status on our respective sides of the project, brainstorm new ideas, and

confront challenges. We also meet with the ECE department and OWL's representatives weekly in an online meeting to discuss our status on the project. Our project progress is version-controlled on a GitHub repository, which is useful for peer code review.

In conclusion, our senior design project is a solution to OWL Integration's open-ended problem statement. The project involves incorporating data-collecting sensors and anomaly detection onto UConn Formula SAE electric racecar using OWL's IoT related products, including the Quackerboard and the DMS, in addition to Google's Coral Board. Our team's deliverables include the machine learning model to detect anomalies in the racecar's battery, and the web dashboard to view the incoming data from the racecar in real-time.

Work Cited

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